GNU Linux Targeted Tools for Arm 64-bit Embedded Processors

NXP GCC for Arm Release 15-May-2020

Release description

Current release is based on:

- gcc 9.2.0
- binutils 2.32
- glibc 2.29
- gdb 8.1.1

This distribution provides tools for 32-bit host system. For 64 bit system, 32 bit libraries are required to run the tools.

Version identification

The version information (emitted via `--version` or `-v`) output format has changed to precisely define a build **BLD** identifier as an integer value whose value will always increase with each subsequent release. Example:

```
$ aarch64-linux-gnu-gcc --version
aarch64-linux-gnu-gcc (BLD = 1649) 9.2.0 20190812 (build.sh rev=gaf57174 s=F920 \
-i /opt/freescale Xarmv8 -V release_gaf57174_build_Fed_Xarmv8)
```

```
$ aarch64-linux-gnu-gcc -v
<snip>
gcc version 9.2.0 20190812 (build.sh rev=gaf57174 s=F920 -i /opt/freescale \
Xarmv8 -V release_gaf57174_build_Fed_Xarmv8) (BLD = 1649)
```

GCC Section Pragmas

GCC supports `#pragma GCC section (text|data|bss|rodata) ".name"` pragma.

This pragma allows to redefine standard sections by user defined section

```
#pragma GCC section (text|data|bss|rodata) ".name"
```

This pragma restores standard section

```
#pragma GCC section (text|data|bss|rodata) "default"
```

Fore example:

```
#pragma GCC section bss ".foobss"
#pragma GCC section data ".foodata"
#pragma GCC section rodata ".foorodata"
#pragma GCC section text ".footext"
char c1;
char c2 = 2;
const char c3 = 3;
int f1 (int x) { return x; }
#pragma GCC section bss "default"
#pragma GCC section data "default"
#pragma GCC section rodata "default"
#pragma GCC section text "default"
```

Installing executables on Windows

Some executables built by MinGW requires additional libraries. Thus to support out-of-the-box execution, libwinpthread.dll comes with toolchain. This library can be moved in appropitate place if host system requires. Or MinGW package can be installed to get executables working.

Installing executables on Linux

"32-bit compatibility libraries" should be installed just in order to run 32-bit toolchains:

Ubuntu 14

```
$ sudo apt-get install lib32z1 lib32ncurses5 lib32bz2-1.0
```

Ubuntu 16

```
$ sudo dpkg --add-architecture i386
$ sudo apt-get update
$ sudo apt-get install libc6:i386 libncurses5:i386 libstdc++6:i386 lib32z1
```

Debian

```
$ sudo apt-get install lib32z1 lib32ncurses5 lib32stdc++6
```

CentOS

```
$ sudo yum install glibc.i686 ncurses-devel.i686
```

GDB with Python

Additionally if you want to use gdb python build (aarch64-linux-gnu-gdb-py) on Linux, you need to install 32 bit python2.7 and compatibility libraries.

Ubuntu

```
$ sudo apt-get install libpython2.7:i386
$ sudo apt-get install python2.7-minimal:i386 python2.7:i386 python2.7-dev:i386
```

And after that switch default python to new one:

```
$ sudo ln -s usr/bin/python2.7 python
```

CentOS

```
$ sudo yum install python-libs.i686
```

Windows

To use gdb python build (aarch64-linux-gnu-gdb-py.exe) on Windows, you need to install 32 bit python2.7 no matter 32 or 64 bit Windows. Please get the package from https://www.python.org/download/.

Release history

Date	Description
29-January-2020	Initial release
bld=1649 rev=gaf57174	No changes